Contact

www.linkedin.com/in/lawp-tech (LinkedIn)

Top Skills

Cursor AI

Retrieval-Augmented Generation (RAG)

Docker

Lawrence P.

Senior Software Engineer | Full-Stack Development | AI/ML & Data Engineering

United States

Summary

I'm a Senior Software Engineer with 10+ years of experience building scalable software solutions across Full-Stack Development, Artificial Intelligence (AI), Machine Learning (ML), and Data Engineering.My expertise includes Python, TypeScript, JavaScript, React, Next.js, Node.js, FastAPI, and Java, with a strong background in backend development, distributed systems, microservices, and cloud-native architecture.I specialize in developing Generative AI applications, Large Language Models (LLMs), AI Agents, Retrieval-Augmented Generation (RAG), and ML-powered solutions, along with data pipelines, API integrations, and automation.I have hands-on experience with AWS, PostgreSQL, SQL, Docker, Kubernetes, and CI/CD, delivering reliable, high-performance applications from concept to production.I'm passionate about solving complex engineering challenges, exploring emerging AI technologies, and building products that create real business impact.

Experience

Robin AI

Senior Software Engineer | Full-Stack & AI

April 2024 - Present (2 years 7 months)

San Francisco Bay Area

As a Senior Software Engineer at Robin AI, I design and build scalable, AI-powered enterprise applications, focusing on full-stack development, Agentic AI, Generative AI, machine learning, and cloud-native architecture.

• Designed and deployed production-grade Agentic AI systems using Anthropic Claude, Amazon Bedrock, AgentCore, and LangChain, implementing multi-agent orchestration, tool calling, and automated contract analysis workflows.
• Built Retrieval-Augmented Generation (RAG) pipelines using Amazon Bedrock Knowledge Bases, OpenSearch, PostgreSQL (pgvector), and vector embeddings, improving document retrieval accuracy and contextual AI responses.

• Developed end-to-end full-stack applications using Python, FastAPI, TypeScript, React, and AWS, building scalable REST APIs, reusable UI components, and enterprise workflow automation.
• Engineered data processing and ML pipelines using AWS Lambda, S3, SQS, DynamoDB, and EventBridge, supporting document ingestion, text extraction, embedding generation, and semantic search.
• Implemented LLM evaluation, prompt engineering, model testing, and AI observability to improve response quality, reliability, and performance across production AI applications.
• Leveraged Claude, Claude Skills, Cursor, and AWS Kiro for AI-assisted development, spec-driven engineering, code generation, automated testing, and developer productivity improvements.
• Designed and maintained containerized microservices using Docker, Kubernetes, Terraform, and CI/CD pipelines, improving application scalability, deployment reliability, and system performance.
• Mentored junior engineers, conducted code reviews, contributed to system architecture decisions, and collaborated with product and engineering teams to deliver reliable, production-ready solutions.

Key Technologies: Python, TypeScript, JavaScript, React, Next.js, FastAPI, AWS, Generative AI, Agentic AI, Claude Skills, Amazon Bedrock, AgentCore, Kiro, Cursor, LangChain, RAG, LLMs.

Electronic Arts (EA)
Software Engineer | AI/ML & Developer Tools
June 2021 - March 2024 (2 years 10 months)
Redwood City, California, United States

At Electronic Arts (EA), I focused on building scalable full-stack applications, AI/ML-powered automation tools, and data engineering solutions to improve internal workflows, marketing operations, and developer productivity.

• Developed internal full-stack platforms using React, TypeScript, Node.js, Python, and PostgreSQL, supporting campaign management, content approvals, and workflow automation.
• Built scalable ETL and data processing pipelines using Python, Pandas, SQL, and AWS S3, transforming campaign metadata, application logs, and operational data into actionable insights.
• Developed machine learning solutions using Python and scikit-learn, implementing classification, feature engineering, and automated content tagging to improve data quality and discoverability.

• Integrated Generative AI and Large Language Models (LLMs) into internal tools during 2023–2024, leveraging Anthropic Claude, Amazon Bedrock, and prompt engineering for content generation and automation.
• Prototyped Retrieval-Augmented Generation (RAG) applications using LangChain, Amazon Bedrock Knowledge Bases, and OpenSearch, enabling semantic search and contextual knowledge retrieval.
• Designed backend services and event-driven workflows using FastAPI, AWS Lambda, SQS, and DynamoDB, supporting asynchronous processing, API integrations, and scalable application architecture.
• Containerized and deployed applications using Docker, AWS ECS, and GitHub Actions, implementing CI/CD pipelines, automated testing, and CloudWatch monitoring.
• Collaborated with engineering, product, and marketing teams while mentoring developers, conducting code reviews, and contributing to software architecture and technical improvements.

Core Technologies: Python, TypeScript, JavaScript, React, Node.js, FastAPI, SQL, PostgreSQL, Pandas, scikit-learn, Machine Learning, Generative AI, LLMs, Claude, Amazon Bedrock, RAG, LangChain, OpenSearch, AWS Lambda, S3, SQS, DynamoDB, ECS, Docker, GitHub Actions, CI/CD.

Cota Healthcare
Software Engineer | Data Engineering & Full-Stack
November 2019 - February 2021 (1 year 4 months)

• Built Python ETL services to ingest oncology EHR extracts through Mirth Connect, normalize diagnosis and treatment fields, and load de-identified records into PostgreSQL for analysis.
• Developed BigQuery datasets and SQL transformations on Google Cloud to consolidate longitudinal patient records, enabling consistent analysis of treatment patterns and clinical outcomes.
• Implemented full-stack analytics features using Angular, TypeScript, Cube.js, and Plotly, allowing researchers to filter patient cohorts and explore oncology trends through interactive dashboards.
• Designed REST APIs and Cube data models on Google Compute Engine, using pre-aggregations and query caching to improve responsiveness for complex clinical analytics workloads.
• Automated clinical data validation with Python, SQL, and scheduled quality checks, flagging missing fields, inconsistent codes, and duplicate records before downstream reporting.

• Improved secure data pipelines using PostgreSQL, Flyway migrations, and access controls, helping maintain traceable schema changes and reliable handling of de-identified healthcare datasets.
• Containerized backend services with Docker and automated CI/CD checks for healthcare data integrations, improving deployment consistency and reducing manual troubleshooting across environments.
• Added cohort filtering by COTA Nodal Address (CNA) in Angular and TypeScript dashboards, helping analysts compare clinically similar oncology patients and examine treatment patterns.

Education

University of Maryland

Bachelor of Science, Computer Science · (April 2014 - October 2019)